Second-Order Linear Homogeneous Equations

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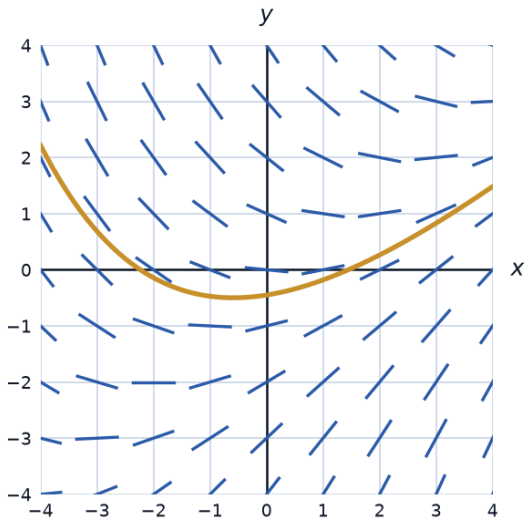
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The auxiliary equation**

Write the auxiliary (characteristic) equation for $y'' - 5y' + 6y = 0$.

2 **[Foundational]** • **Target: The auxiliary equation**

Write and simplify the auxiliary equation for $2y'' + 4y' - 6y = 0$.



3 **[Foundational]** • **Target: The auxiliary equation**

Explain why substituting $y = e^{rx}$ into $ay'' + by' + cy = 0$ produces the auxiliary equation $ar^2 + br + c = 0$.

4 **[Foundational]** • **Target: Distinct real roots**

Solve $y'' - 9y = 0$ (note this factors as a difference of squares in r).

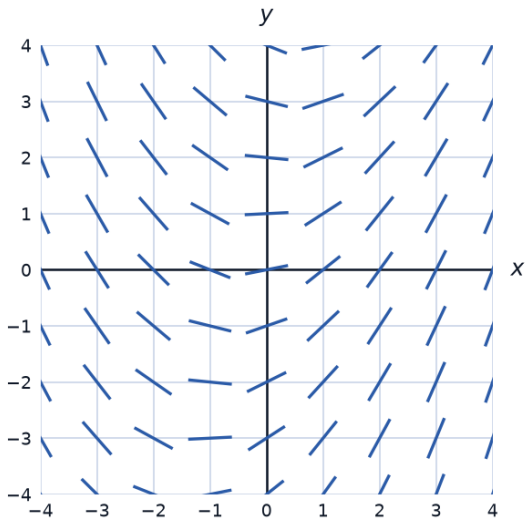
5 **[Foundational]** • **Target: Repeated roots**

Explain why, for a repeated root r, $y_2 = xe^{rx}$ (not just a second copy of e^{rx}) is needed to build the general solution.

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6 [Foundational] • Target: Classifying without fully solving

Without solving, state whether $y'' + 6y' + 5y = 0$ has real, repeated, or complex roots, using the discriminant $b^2 - 4ac$ of the auxiliary equation.



7 [Foundational] • Target: Classifying without fully solving

Without solving, classify the roots of $y'' + 10y' + 25y = 0$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Application: a damped spring

A mass–spring–damper system satisfies $y'' + 4y' + 5y = 0$. Find the general solution and classify the damping (over/critically/under-damped).

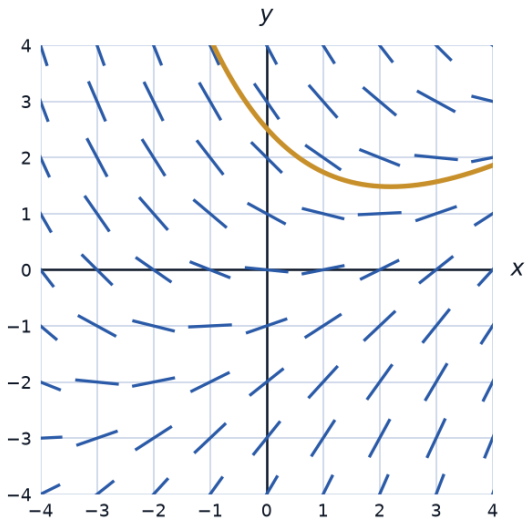
9 [Intermediate] • Target: Application: a damped spring

Solve $y'' + 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.

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10 **[Intermediate]** • **Target: Application: a damped spring**

For the general damped spring equation $my'' + cy' + ky = 0$ $(m, c, k > 0)$, explain in one sentence why the real part of every root of the auxiliary equation is always negative (so every solution decays).



11 **[Intermediate]** • **Target: Verifying general solutions and the superposition principle**

Verify by direct substitution that $y = C_1e^{2x} + C_2e^{3x}$ solves $y'' - 5y' + 6y = 0$ for any constants C_1, C_2.

12 **[Intermediate]** • **Target: Verifying general solutions and the superposition principle**

State the superposition principle for a linear homogeneous ODE, and explain why it guarantees that $y = C_1y_1 + C_2y_2$ solves $ay'' + by' + cy = 0$ whenever y_1, y_2 each do.

13 **[Intermediate]** • **Target: Verifying general solutions and the superposition principle**

Find the auxiliary equation's roots for $y'' - 4y = 0$, then verify that both $y = e^{2x}$ and $y = e^{-2x}$ individually satisfy the ODE.

14 **[Intermediate]** • **Target: Boundary value problems**

Solve the boundary value problem $y'' + y = 0$, $y(0) = 0$, $y(\pi/2) = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

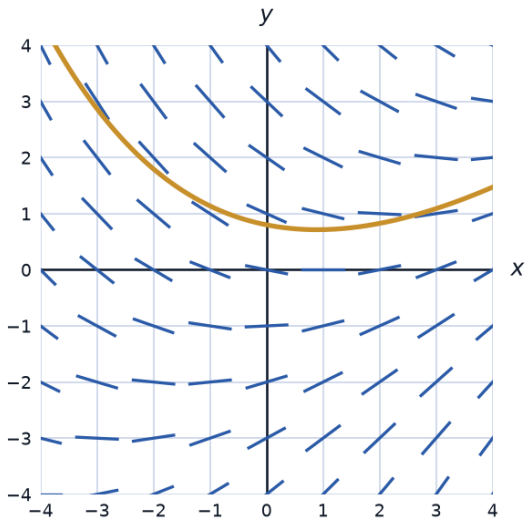
15 **[Challenging]** • **Target: Boundary value problems** • **Cross-Lesson: 1.01 Separable Equations**

Explain why the boundary value problem $y'' + 4y = 0$, $y(0) = 0$, $y(\pi) = 1$ has NO solution, while $y(0) = 0$, $y(\pi/4) = 1$ does.

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16 **[Challenging]** • **Target: Boundary value problems** • **Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor**

Find the indicial equation and roots about $x = 0$ for the differential equation $2x^2y'' - xy' + (x - 5)y = 0$, and state the form of the two linearly independent Frobenius solutions.



17 **[Challenging]** • **Target: The auxiliary equation** • **Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients**

Find the general solution of the differential equation $y'' + y = 4\sin x + 2e^{-x}$.

18 **[Challenging]** • **Target: Distinct real roots** • **Cross-Lesson: 5.01 Variation of Parameters**

Find the general real solution of $\mathbf{x}' = (-2,\ -5; 1, 0)\mathbf{x}$, classify the origin, and determine whether trajectories rotate clockwise or counterclockwise.

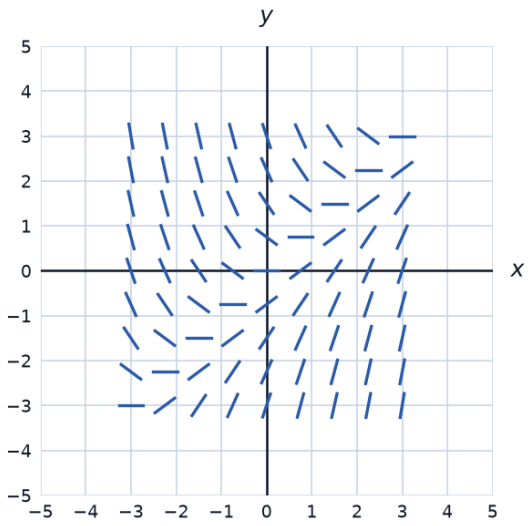
19 **[Challenging]** • **Target: Repeated roots** • **Cross-Lesson: 1.01 Separable Equations**

[Systems of Linear Differential Equations, Phase Plane Topology & Matrix Exponentials · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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20 [Challenging] • Target: Complex roots • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Solve the double-resonance equation $y'' - 6y' + 9y = 4e^{3t}$ with $y(0) = 1$ and $y'(0) = 3$.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The auxiliary equation

Consider the second-order linear homogeneous differential equation $y'' - 2y' + ky = 0$. We are given that $y_1(x) = e^{3x}$ is a solution to this equation. \\ (a) Find the value of the parameter k. \\ (b) Find the general solution to the differential equation.

a. $k = -3, y(x) = c_1e^{3x} + c_2e^{-x}$

b. $k = 3, y(x) = c_1e^{3x} + c_2xe^{3x}$

c. $k = -3, y(x) = c_1e^{3x} + c_2e^{x}$

d. $k = 9, y(x) = c_1e^{3x} + c_2e^{-3x}$

22 [Foundational] • Target: Distinct real roots

Consider the second-order linear homogeneous differential equation $y'' - 3y' + 2y = 0$. Find the general solution to this equation.

a. $y(x) = c_1e^{x} + c_2e^{2x}$

b. $y(x) = c_1e^{-x} + c_2e^{-2x}$

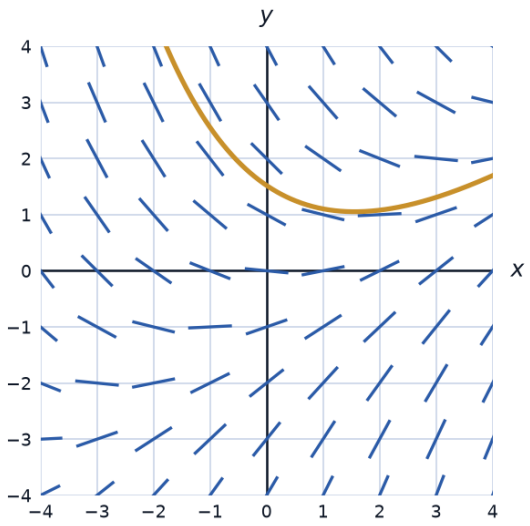
c. $y(x) = c_1\cos(x) + c_2\sin(x)$

d. $y(x) = c_1xe^{x} + c_2e^{x}$

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23 [Foundational] • Target: Repeated roots

Consider the second-order linear homogeneous differential equation $y'' + P(x)y' + Q(x)y = 0$. For the equation $x^2y'' + xy' + (x^2 - u^2)y = 0$ (Bessel's equation of order u), the point $x = 0$ is a regular singular point. If we assume a series solution of the form $y(x) = \sum_{n=0}^{\infty} a_n x^{n+r}$, where $a_0 eq 0$, the indicial equation is given by $r^2 - u^2 = 0$. For this problem, let $u = 1$, so the indicial equation is $r^2 - 1 = 0$. This yields two distinct roots, $r_1 = 1$ and $r_2 = -1$. For the larger root $r_1 = 1$, the power series solution takes the form $y_1(x) = x + \sum_{n=1}^{\infty} a_n x^{n+1}$. Substituting this into Bessel's equation of order 1, and after considerable algebraic manipulation, the recurrence relation for the coefficients a_n is found to be $a_n = -\frac{a_{n-2}}{n(n+2)}$ for $n \geq 2$. Given $a_0 = 1$, find the value of a_3 for the solution corresponding to the larger root $r_1 = 1$.



a. 0

b. 1/24

c. 1/48

d. 1/12

24 [Foundational] • Target: Complex roots

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

a. 5

b. -5

c. 6

d. -6

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25 [Foundational] • Target: Classifying without fully solving

Consider the second-order linear homogeneous differential equation $y''(x) + 5y'(x) + 6y(x) = 0$. If the auxiliary equation has roots r_1 and r_2, what is the value of $r_1 + r_2$?

a. 5

b. -5

c. 6

d. -6

26 [Foundational] • Target: Application: a damped spring

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the characteristic equation has roots r_1 and r_2 with $r_1 < r_2$, what is the value of r_1?

a. 1

b. 2

c. 3

d. 6

27 [Foundational] • Target: Verifying general solutions and the superposition principle

For which values of k does every nonzero solution of $y'' + 4y' + ky = 0$ genuinely oscillate (the auxiliary equation has non-real roots)?

a. $k < 4$

b. $k = 4$

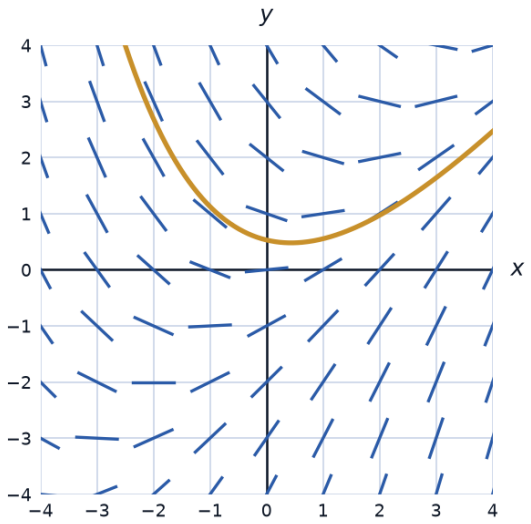
c. $k > 0$

d. $k > 4$

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28 [Foundational] • Target: Boundary value problems

Consider the second-order linear homogeneous differential equation: $x^2y'' + xy' + (x^2 - u^2)y = 0$, where u is a constant. This is Bessel's equation of order u. Assume $u = 1/2$ and that $x > 0$. x is a regular singular point. Determine the indicial equation for this differential equation and find its roots.



a. Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{2}$

b. Indicial equation: $r^2 - r + \frac{1}{4} = 0$; roots: $r = \frac{1}{2}, \frac{1}{2}$

c. Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{4}$

d. Indicial equation: $r^2 + \frac{1}{4} = 0$; roots: $r = \pm\frac{i}{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The auxiliary equation

Consider the second-order linear homogeneous differential equation: $y'' - xy' + y = 0$. The point $x = 0$ is an ordinary point. Find the first two non-zero terms of the power series solution around $x = 0$, assuming $y(0) = 1$ and $y'(0) = 0$.

a. $1 + \frac{1}{2}x^2 + \ldots$

b. $1 + \frac{1}{3}x^3 + \ldots$

c. $1 - \frac{1}{2}x^2 + \ldots$

d. $1 - \frac{1}{3}x^3 + \ldots$

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30 **[Intermediate]** • **Target: Distinct real roots**

Consider the nonhomogeneous second-order linear differential equation $y'' - y = \frac{1}{1+e^x}$. If $y_1(x) = e^x$ and $y_2(x) = e^{-x}$ are two linearly independent solutions to the associated homogeneous equation, find the Wronskian of y_1 and y_2.

a. 2

b. -2

c. 0

d. $e^x - e^{-x}$

31 **[Intermediate]** • **Target: Repeated roots**

A damped spring satisfies $y'' + 2y' + 10y = 0$, $y(0) = 0$, $y'(0) = 6$. Find $y(\pi/6)$.

a. $2e^{\pi/6}$

b. $2e^{-\pi/6}$

c. $-2e^{-\pi/6}$

d. $e^{-\pi/6}$

32 **[Intermediate]** • **Target: Complex roots**

Consider the second-order linear non-homogeneous differential equation $y'' + 4y = f(t)$, where $f(t)$ is a piecewise constant function defined as $f(t) = \{2, , 0 \leq t < 1; 0, , t \geq 1$. The Laplace transform of $f(t)$ can be written in the form $F(s) = \frac{2}{s} - \frac{2e^{-s}}{s}$. Find the Laplace transform of the differential equation, $s^2Y(s) - sy(0) - y'(0) + 4Y(s) = F(s)$, given initial conditions $y(0) = 1$ and $y'(0) = 0$. What is the expression for $Y(s)$?

a. $Y(s) = \frac{1}{s-1} + \frac{2}{s(s^2+4)} - \frac{2e^{-s}}{s(s^2+4)}$

b. $Y(s) = \frac{s}{s^2+4} + \frac{2}{s(s^2+4)} - \frac{2e^{-s}}{s(s^2+4)}$

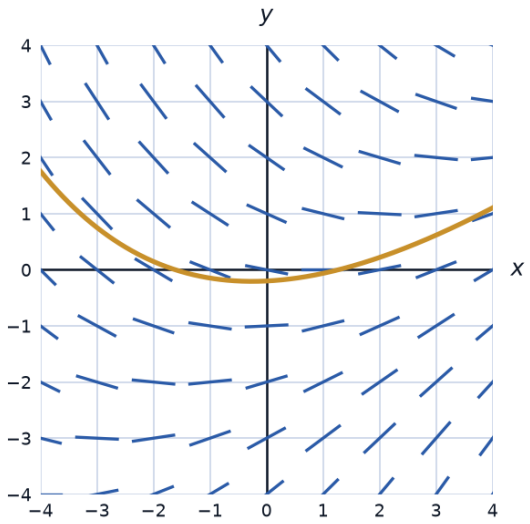
c. $Y(s) = \frac{1}{s^2+4} + \frac{2}{s(s^2+4)} - \frac{2e^{-s}}{s(s^2+4)}$

d. $Y(s) = \frac{s}{s^2+4} + \frac{2}{s} - \frac{2e^{-s}}{s}$

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33 **[Intermediate]** • **Target: Classifying without fully solving**

Consider the nonhomogeneous second-order linear differential equation $y'' + 4y = 8$. What is the particular solution, $y_p(x)$, to this equation?



a. 2

b. $2x$

c. $2\sin(2x)$

d. $2\cos(2x)$

34 **[Intermediate]** • **Target: Application: a damped spring**

Consider the nonhomogeneous linear second-order differential equation $y'' + y = 5$. If $y_p(x) = A$ is a particular solution, what is the value of A?

a. 1

b. 5

c. 1/5

d. 0

35 **[Intermediate]** • **Target: Verifying general solutions and the superposition principle**

Consider the second-order linear nonhomogeneous differential equation: $y''(x) - 3y'(x) + 2y(x) = 4$. Find the particular solution, $y_p(x)$, using the method of Undetermined Coefficients.

a. $y_p(x) = 2$

b. $y_p(x) = 2x$

c. $y_p(x) = 4$

d. $y_p(x) = 4x$

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36 [Intermediate] • Target: Boundary value problems

Consider the second-order linear nonhomogeneous differential equation $y'' - y = \sin x$. If $y_p(x) = A\cos x + B\sin x$ is a particular solution, find the values of the constants A and B.

a. $A = 0, B = \frac{1}{2}$

b. $A = -\frac{1}{2}, B = 0$

c. $A = 0, B = -\frac{1}{2}$

d. $A = \frac{1}{2}, B = 0$

37 [Intermediate] • Target: The auxiliary equation

Find $y(2)$ for the differential equation $y' - y = 4$ satisfying $y(0) = 1$.

a. $5e^2 + 4$

b. $5e^2 - 4$

c. $e^2 - 4$

d. $4e^2 - 5$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Distinct real roots • Cross-Lesson: 5.01 Variation of Parameters

An RC circuit with $R = 20\,\Omega$ and $C = 0.01$ F is driven by $E(t) = 100\sin(10t)$ V. What is the first-order differential equation for the charge $q(t)$?

a. $q' + 10q = 5\sin(10t)$

b. $q' + 5q = 10\sin(10t)$

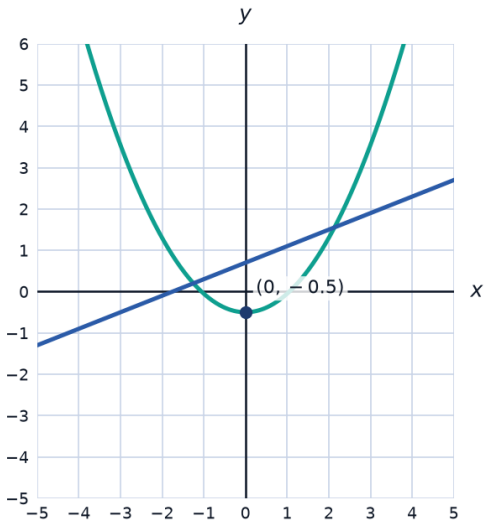
c. $q' + 5q = 5\sin(10t)$

d. $20q' + 5q = \sin(10t)$

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39 [Challenging] • Target: Repeated roots • Cross-Lesson: 1.01 Separable Equations

Second-order ODE with complex roots $r = \alpha \pm \beta i$: general solution includes:



a. $e^{j\beta x}$

b. $C_1 e^{\alpha x} + C_2 e^{\beta x}$

c. $(C_1 + C_2 x)e^{(\alpha + \beta)x}$

d. $e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$

40 [Challenging] • Target: Complex roots • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the differential equation $\frac{dy}{dx} = \frac{\cos(x)}{y^2}$ for $y > 0$ and $x \in [0, \frac{\pi}{2}]$. If $y(0) = 2$, find the value of $y(\frac{\pi}{2})$.

a. $\sqrt[3]{6}$

b. $\sqrt[3]{3}$

c. $\sqrt[3]{9}$

d. $\sqrt[3]{12}$

41 [Challenging] • Target: Classifying without fully solving • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Consider the system of linear differential equations given by: $\begin{cases} \frac{dx}{dt} = -x + 2y; \frac{dy}{dt} = 3x - 2y \end{cases}$ Let the general solution be of the form $\mathbf{x}(t) = c_1 \mathbf{v}_1 e^{\lambda_1 t} + c_2 \mathbf{v}_2 e^{\lambda_2 t}$, where $\lambda_1 < \lambda_2$. Determine the value of λ_2.

a. 1

b. 2

c. 3

d. 4

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42 [Challenging] • Target: Application: a damped spring • Cross-Lesson: 5.01 Variation of Parameters

Consider the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$. If the solution curve passes through the point $(1, 2)$, find the value of the constant of integration, C, when the equation is integrated in the form $y^2 = x^2 + C$.

a. 1

b. 2

c. 3

d. 4

43 [Challenging] • Target: Verifying general solutions and the superposition principle • Cross-Lesson: 1.01 Separable Equations

A tank holds 60 L of pure water. Brine with concentration 2 kg/L enters at 4 L/min, and the thoroughly mixed solution leaves at 2 L/min. What is the salt concentration in the tank after 30 minutes?

a. 1.5 kg/L

b. 1.2 kg/L

c. 1.8 kg/L

d. 2.0 kg/L

44 [Challenging] • Target: Boundary value problems • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Solve $y' + y\sec x = \cos x$ on $(-\pi/2, \pi/2)$ with $y(0) = 0$. What is the value of $y(\pi/3)$?

a. $\frac{2\pi + 3}{6}$

b. $\frac{(2\pi + 3)(2 + \sqrt{3})}{6}$

c. $\frac{(2\pi + 3)(2 - \sqrt{3})}{6}$

d. $\frac{(2\pi - 3)(2 - \sqrt{3})}{6}$

45 [Challenging] • Target: The auxiliary equation • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A tank contains 50 gal of pure water. Brine with 2 lb/gal of salt enters at 3 gal/min, and the well-stirred solution drains out at 1 gal/min. How much salt is in the tank when the volume reaches 100 gallons?

a. $200 - 25\sqrt{2}$ lb

b. 150 lb

c. $200 - 50\sqrt{2}$ lb

d. $200 - 50\sqrt{3}$ lb

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